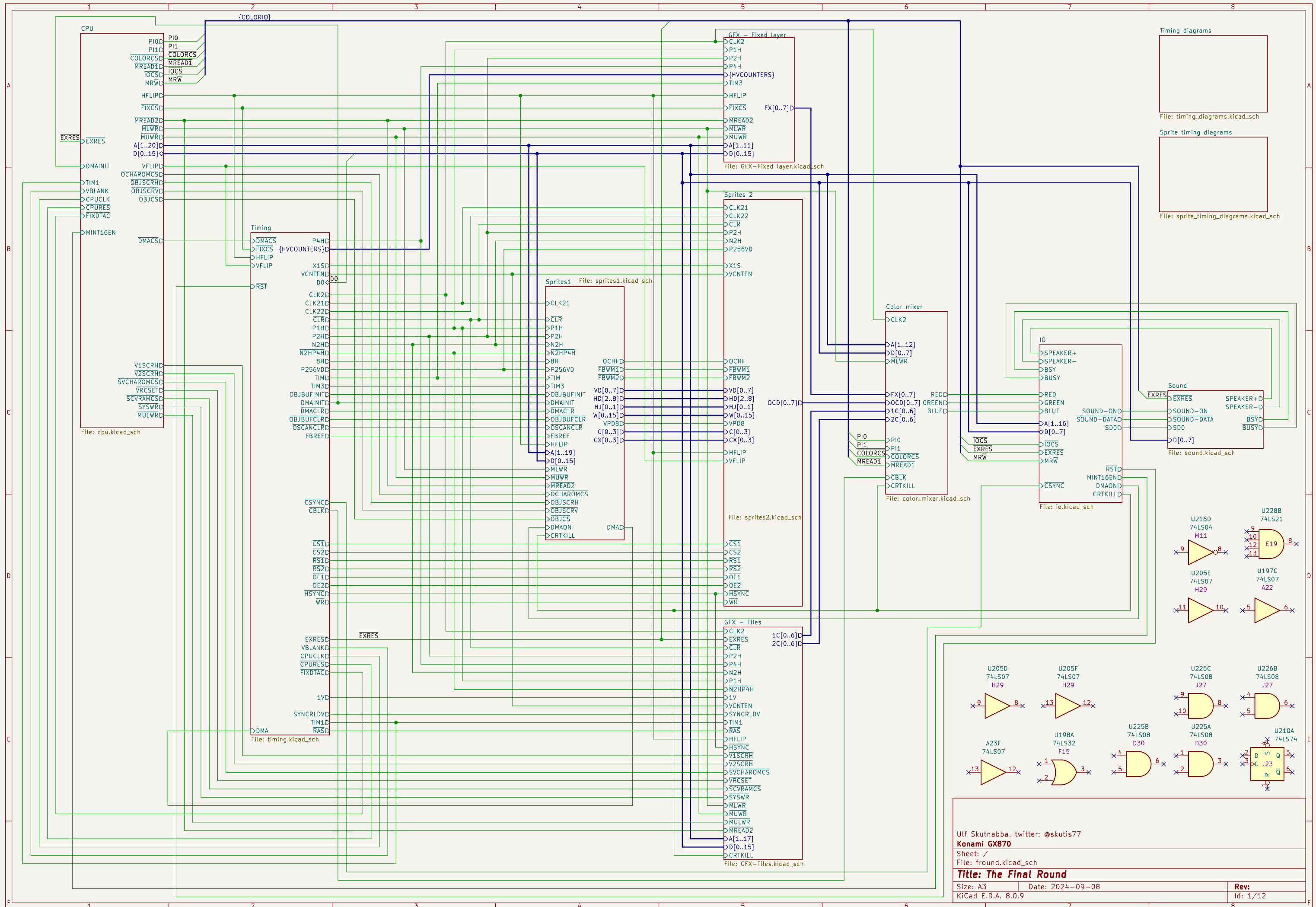
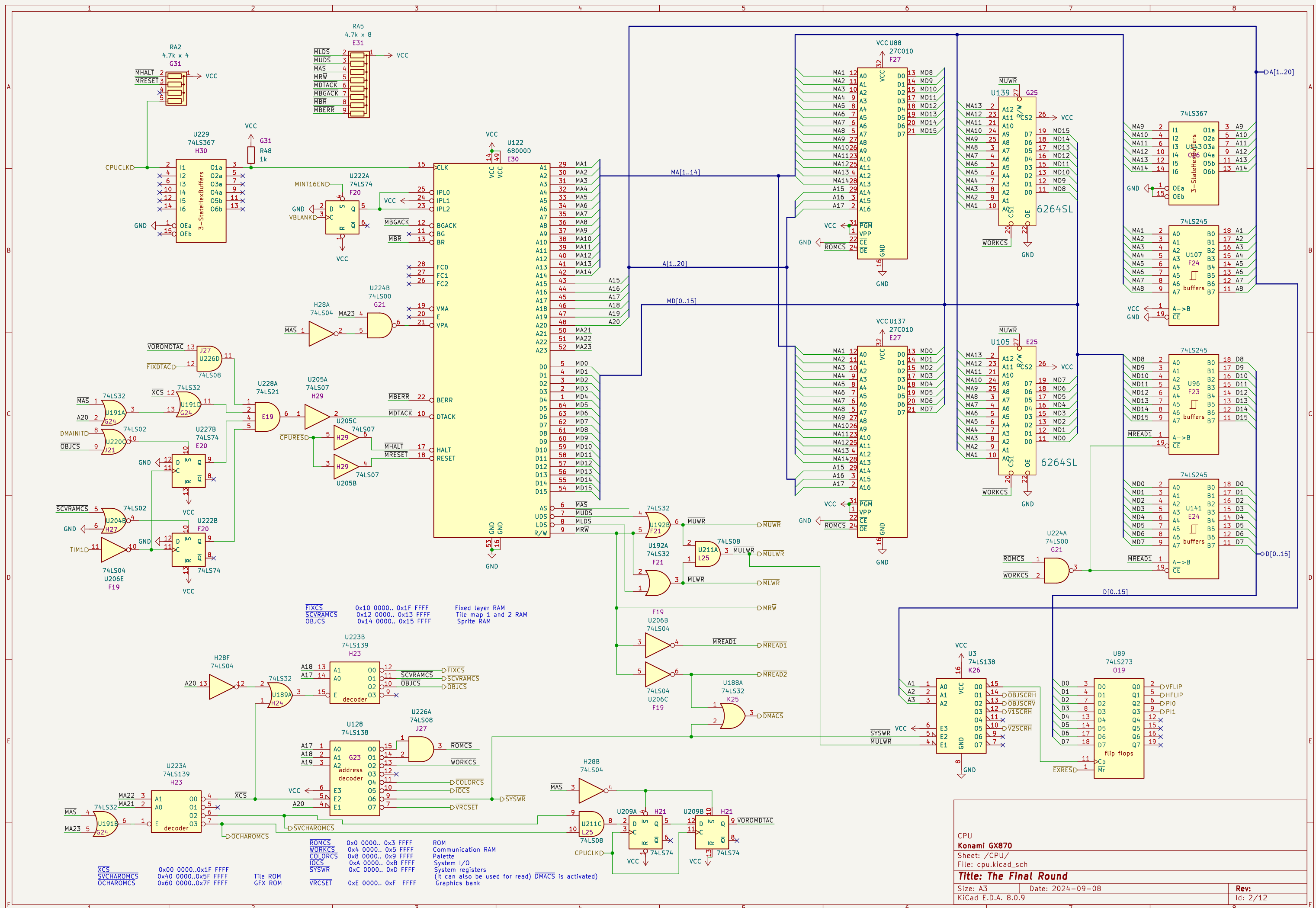
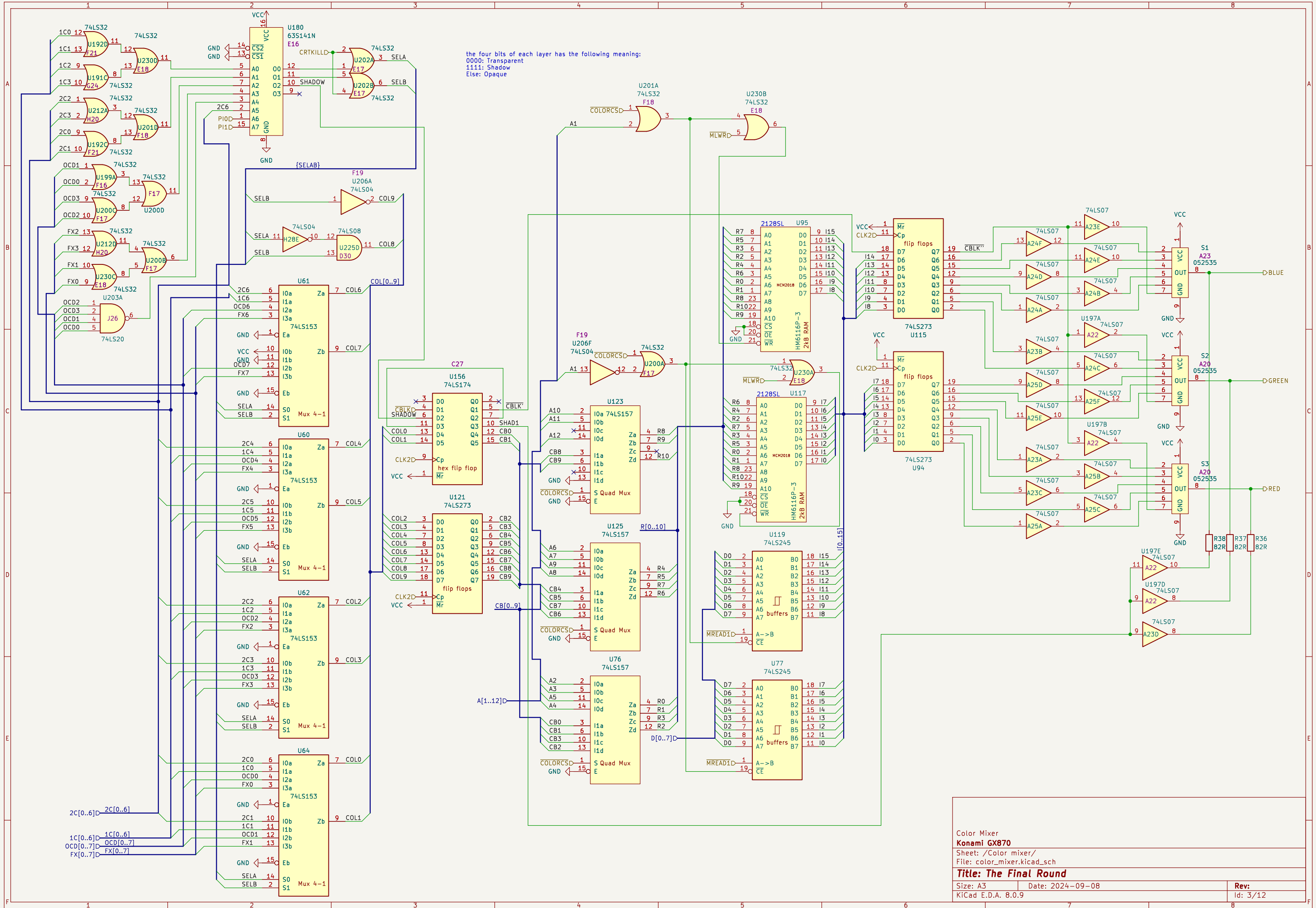
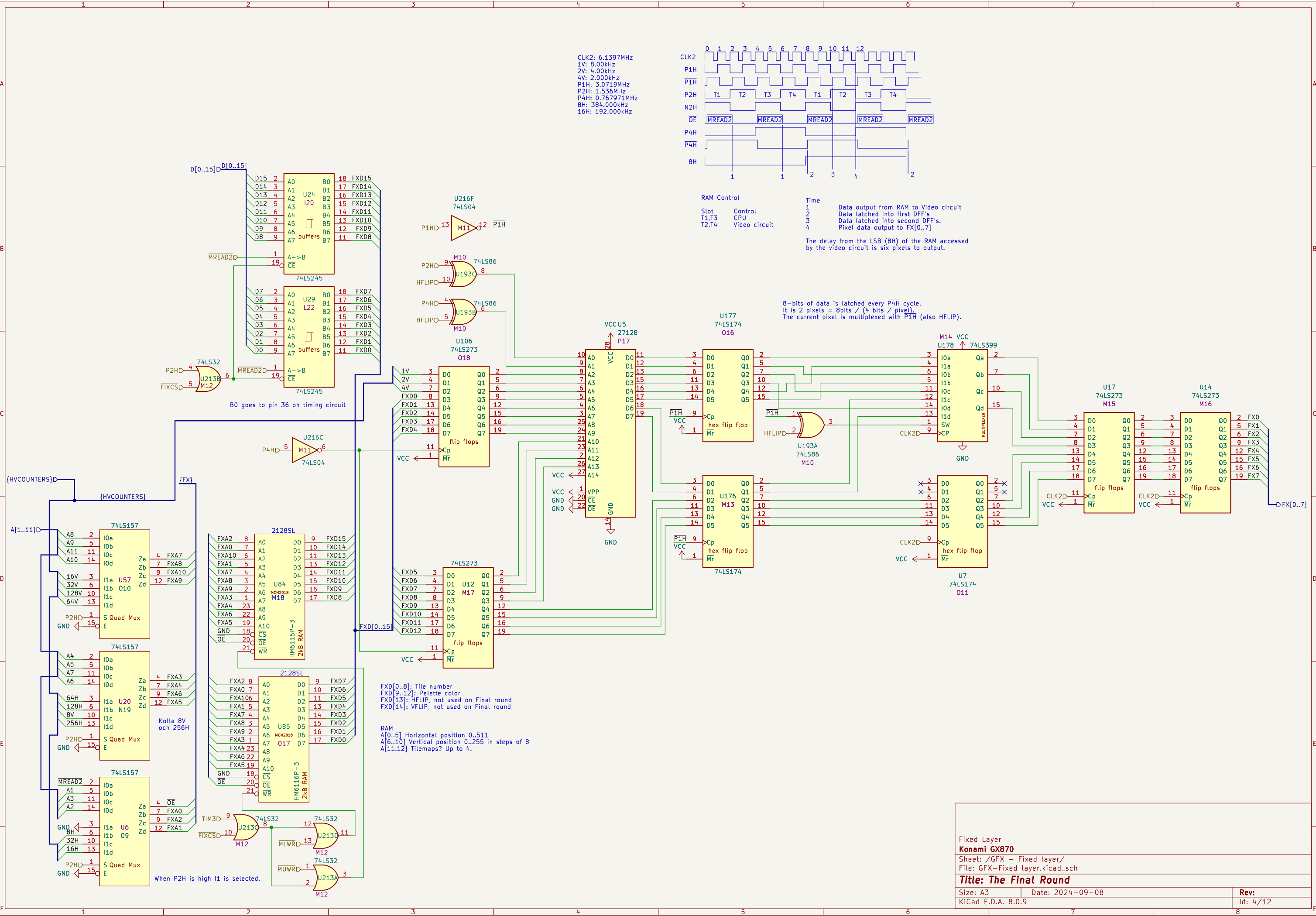
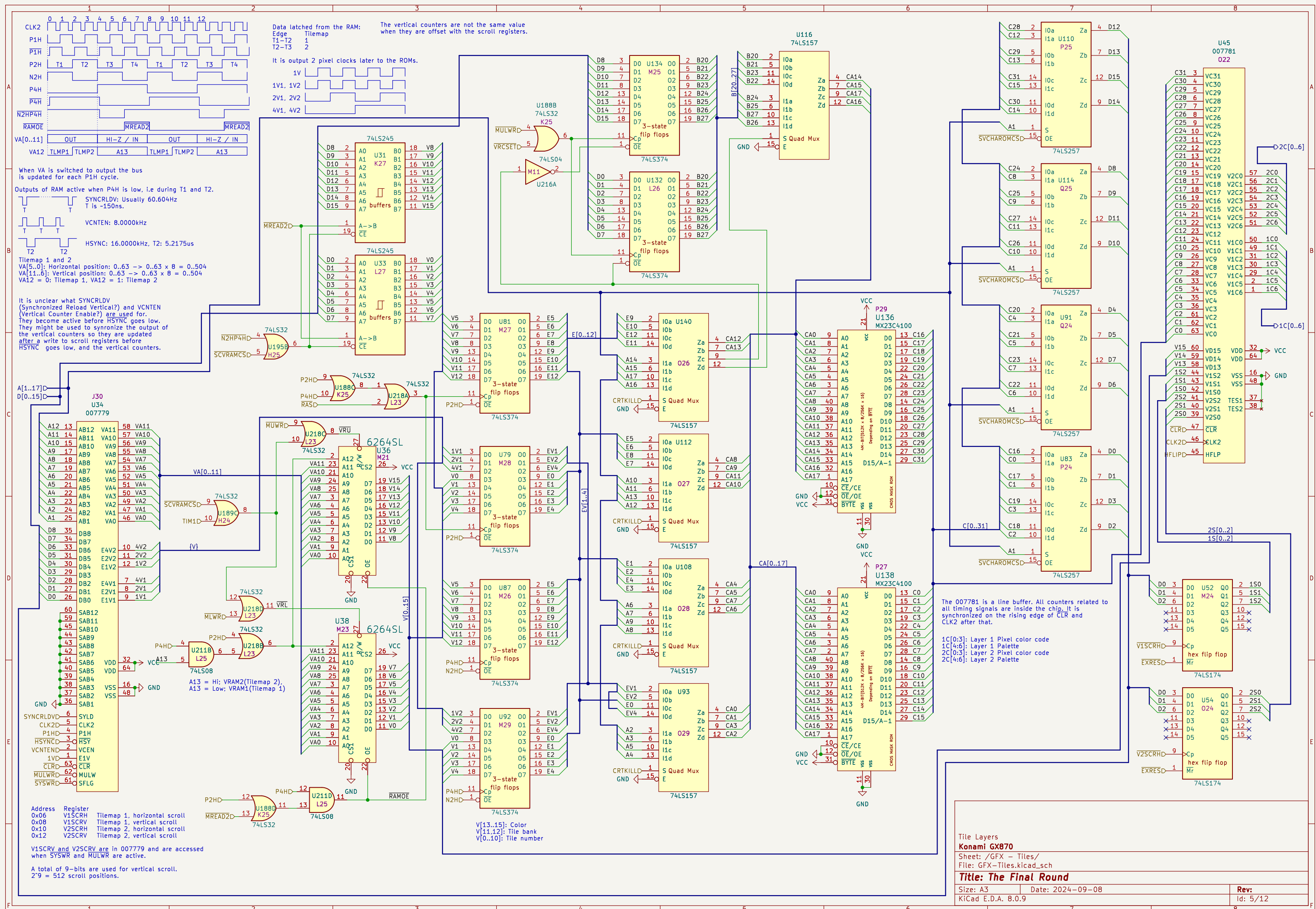


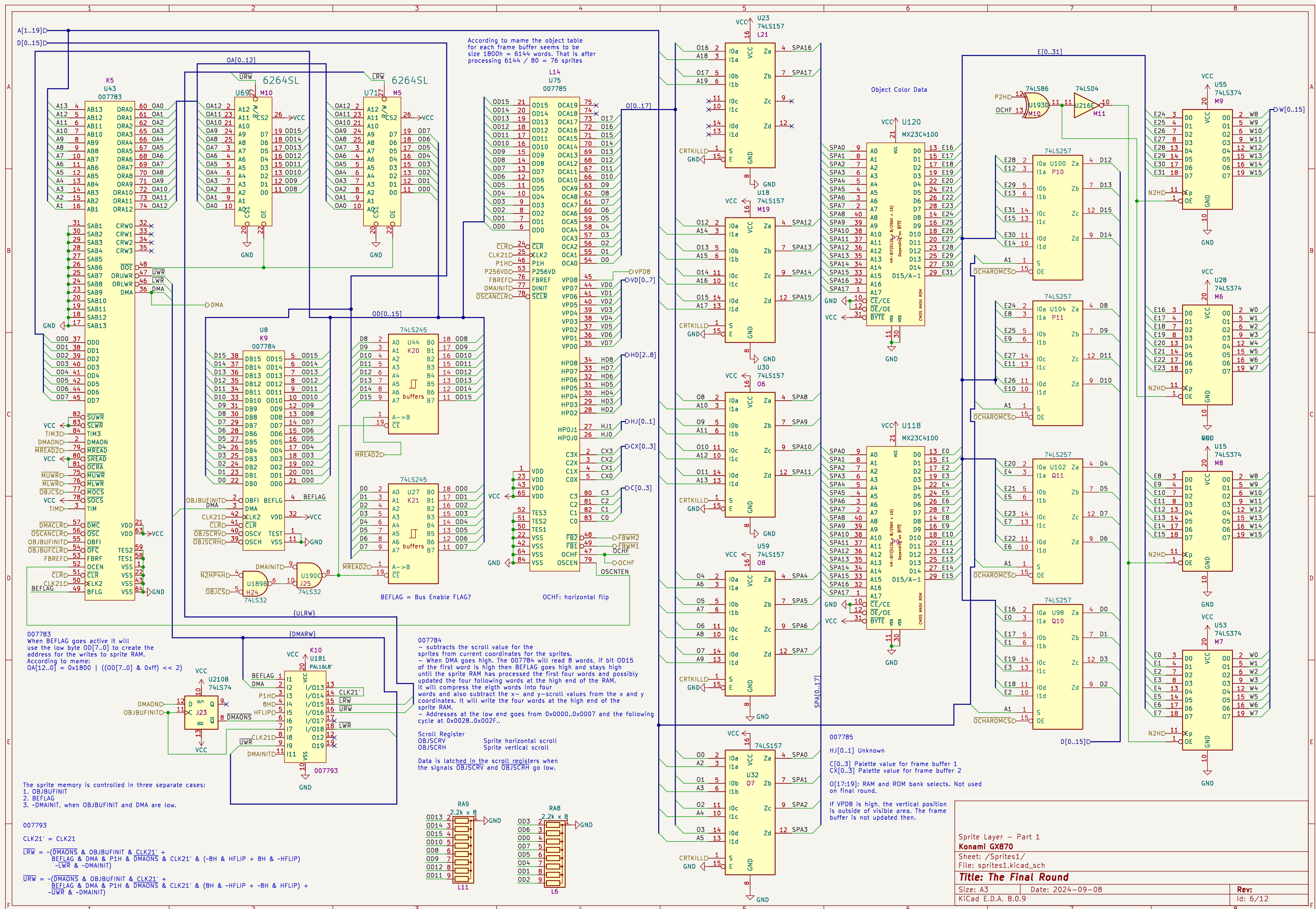


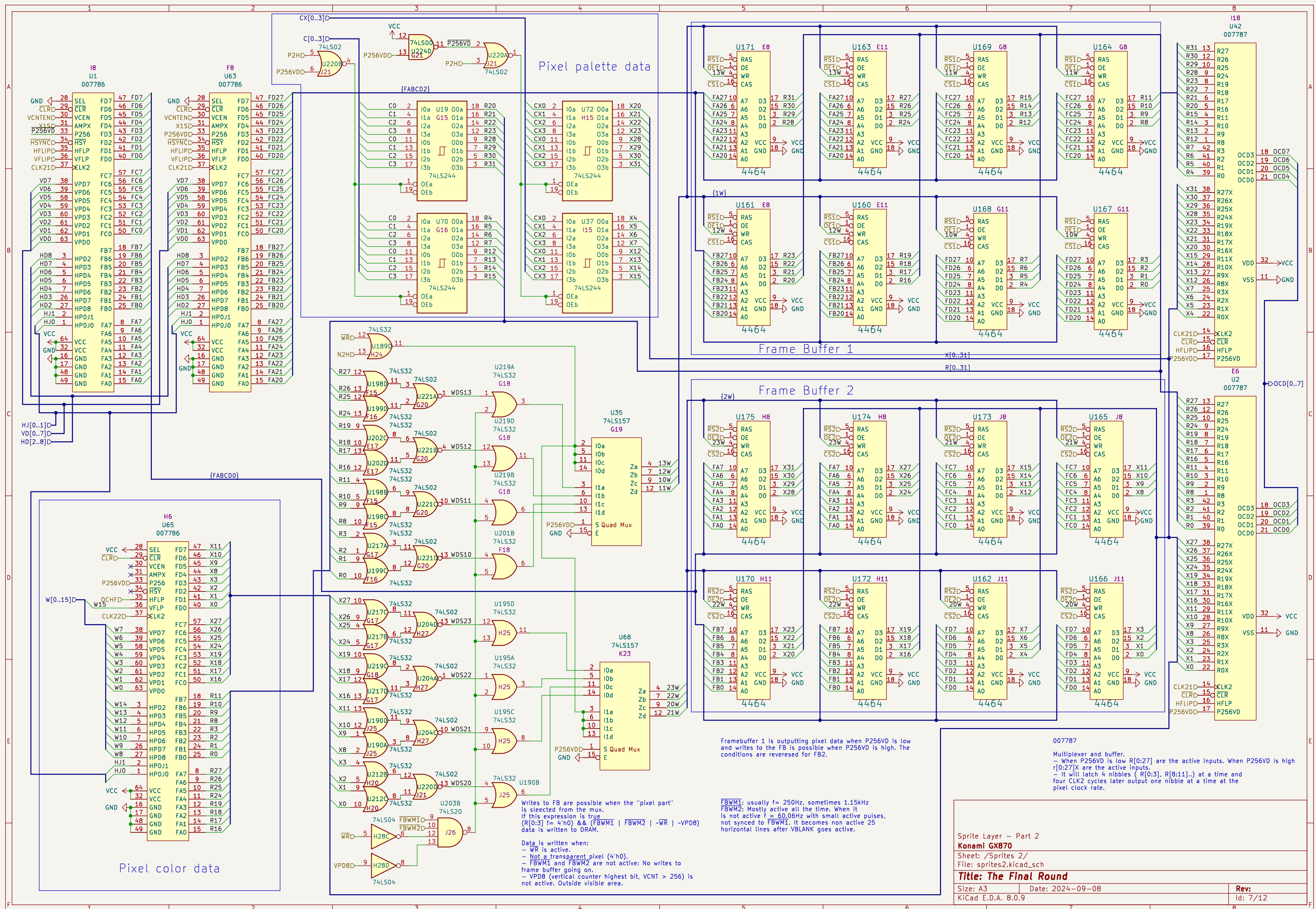




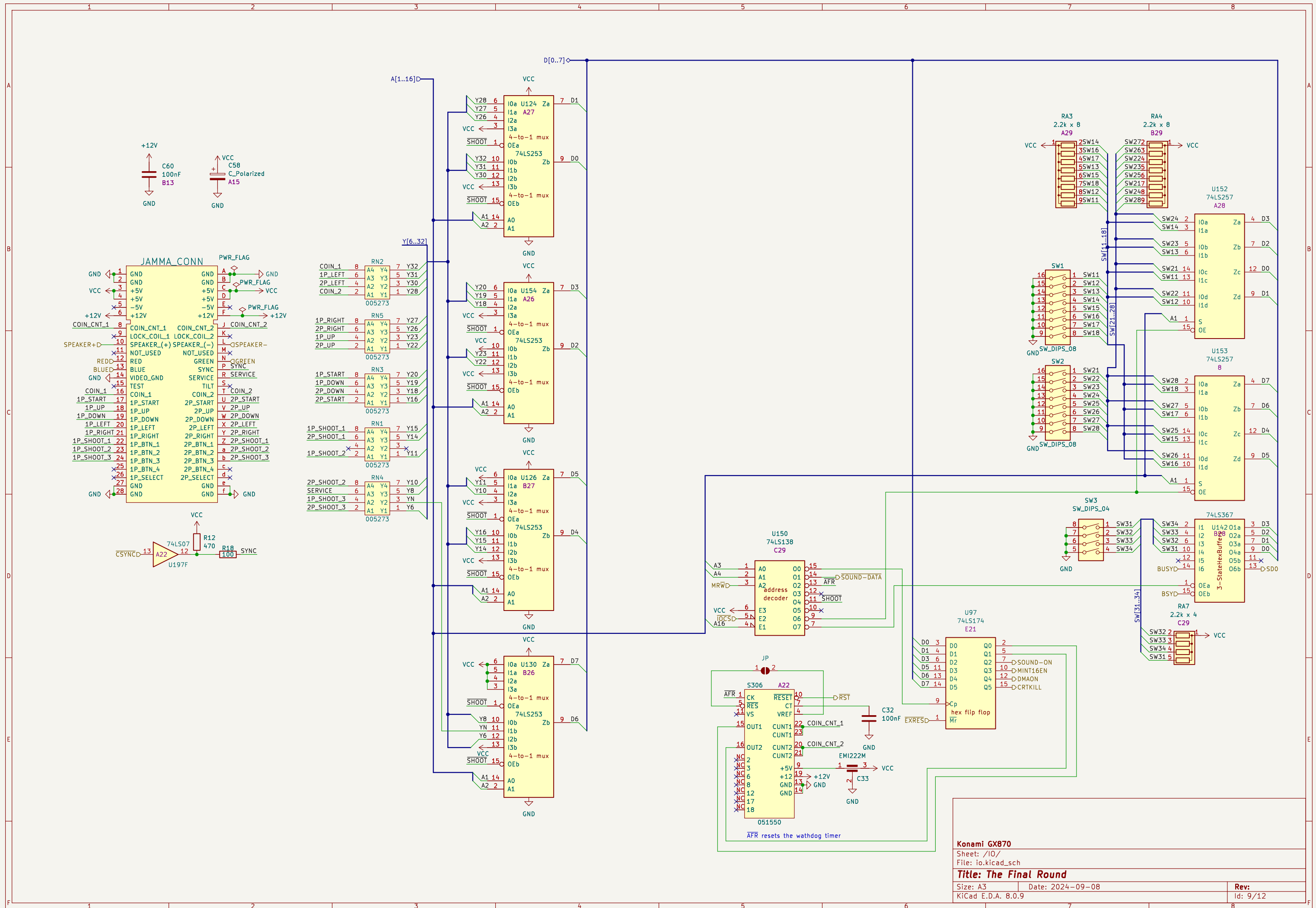












Konami GX870

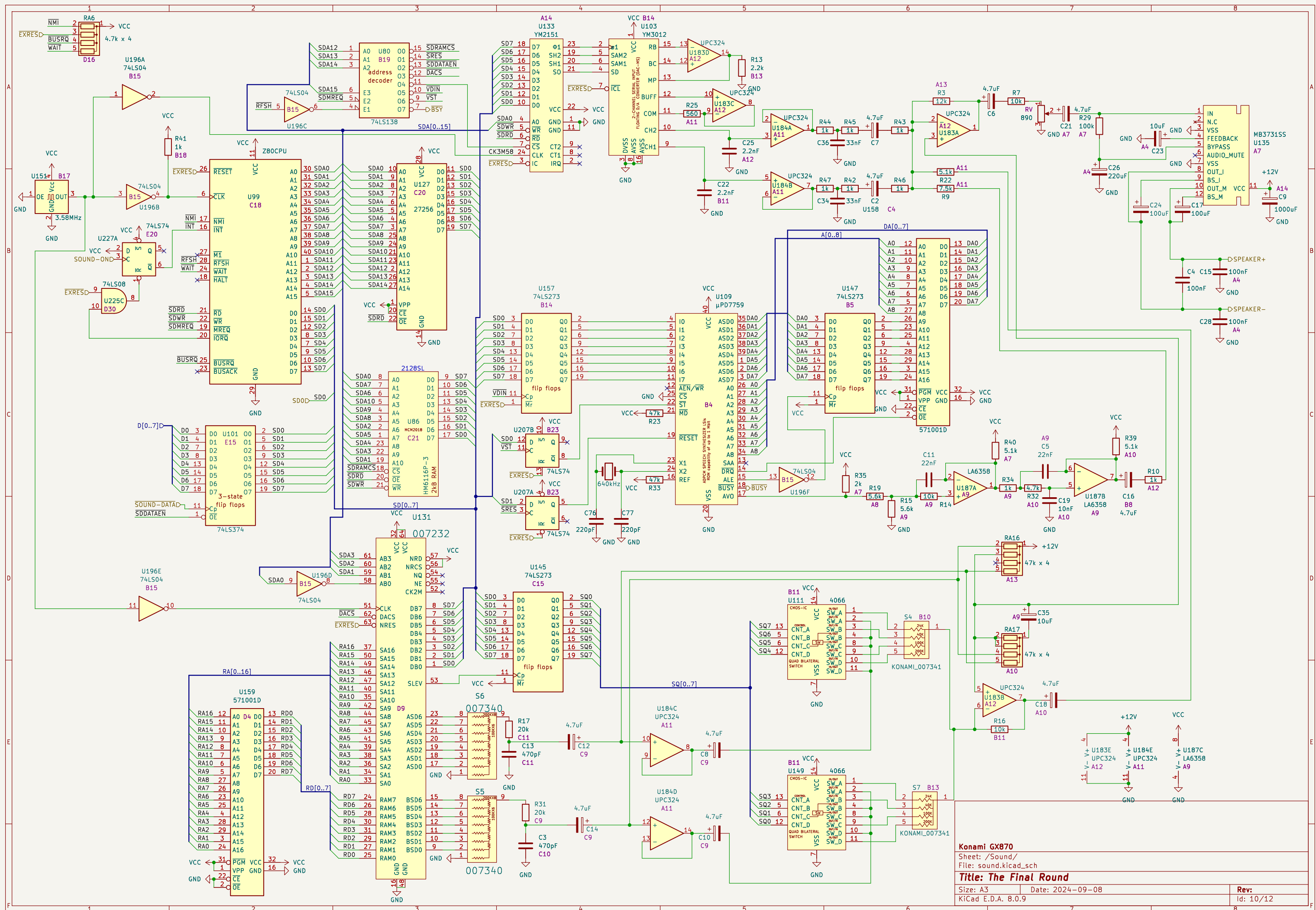
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Title: The Final Round

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Date: 2024-09-08

Rev:
Id: 9/12



1

2

3

4

5

6

7

8

A

B

Horizontal signals

$f = \text{CLK2} / 8 = 768\text{kHz}$

$f = \text{CLK2} / 16 = 384\text{kHz}$

$f = \text{CLK2} / 32 = 192\text{kHz}$

$f = \text{CLK2} / 64 = 96\text{kHz}$

$f = \text{CLK2} / 128 = 48\text{kHz}$

$f = 2/3 \times \text{CLK2} / 128 = 32\text{kHz}$

$f = 2/3 \times \text{CLK2} / 256 = 16\text{kHz}$

$f = 2/3 \times \text{CLK2} / 256 = 16\text{kHz}$

$f = 16\text{kHz} / 2 = 8\text{kHz}$

$f = 16\text{kHz} / 2 = 8\text{kHz}$

$f = 16\text{kHz} / 2 = 8\text{kHz}$

CLR

P4H

8H

16H

32H

64H

128H

256H

HSYNC

HBLK (CBLK)

CSYNC

VCNTEN

1V

The first HSYNC is at HCOUNT 175 after reset.

- The first VCNTEN is skipped after reset. It goes low 140ns before HSYNC goes low, and high again when HSYNC goes low. VCNTEN is active right before every second falling edge of HSYNC.
- CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

- $\overline{\text{CPURES}}$ goes high, and stays high, on the seventh falling edge of $\overline{\text{HSYNC}}$.

Vertical signals

$f = 16\text{kHz} / 2 = 8\text{kHz}$	1V	
$f = 16\text{kHz} / 4 = 4\text{kHz}$	2V	
$f = 16\text{kHz} / 8 = 2\text{kHz}$	4V	

The diagram illustrates the timing of various video signals. The signals shown are:

- CTR**: Clock signal, 16kHz.
- VSYNC**: Vertical sync signal, 60.606Hz.
- VBLANK**: Vertical blanking signal.
- FBREF**: Frame buffer refresh signal.
- SYNCRLDV**: Sync signal ready.
- OBJBUFINIT**: Object buffer init signal.
- OBJBUFCLR**: Object buffer clear signal.
- OSCANCLR**: Oscillator clear signal.
- P256VD**: 256VD signal.

The diagram is divided into two sections: 'C' (Control) and 'D' (Data). The 'C' section shows the timing of the CTR, VSYNC, VBLANK, FBREF, and SYNCRLDV signals. The 'D' section shows the timing of the OBJBUFINIT, OBJBUFCLR, OSCANCLR, and P256VD signals. The diagram includes annotations for the period of the signals and the relationship between them.

[illegible]

The numbers in the $\overline{\text{VSYNC}}$ and $\overline{\text{VBLK}}$ diagrams are $\overline{\text{HSYNC}}$ cycles.
All edges are synchronised to the falling edge of $\overline{\text{HSYNC}}$.

Front Porch (FP) 24 px,
Back Porch (BP) 8 px

Front Porch (FP) and
Back Porch (BP) 16 $\overline{\text{HSYNC}}$ cycles.

Front Porch (FP) and Back Porch (BP) 16 HSYNC cycles.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

$\overline{\text{CBLK}}''$ is at the output of color mixer.

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